Whole-Body Teleoperation with Apple Vision Pro

Internship Progress Report

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1. Purpose of This Report

I joined this lab as a Mitacs GRI summer intern. This report summarizes what I have done over the past two to three months and shares that progress with my supervisor. It is written for someone who may not specialize in robotics: technical terms are kept to a minimum, and abbreviations are explained when they first appear.

The work builds on code and documentation left by a previous graduate student (April–May 2026). My contribution is the extension, integration, tuning, and documentation described below—not a project started from zero.

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2. Problem Statement and Motivation

The problem I set out to address is whether a single Apple Vision Pro headset—without extra controllers or body-worn hardware—can teleoperate an entire Unitree G1 humanoid, including walking, turning, arm motion, and finger control, rather than only one arm or the upper body.

This goal came from three practical needs. First, the prior codebase already connected Vision Pro hand tracking to the robot arms and one Inspire dexterous hand, but whole-body locomotion and a repeatable end-to-end workflow were still missing. Second, the lab uses the SONIC whole-body controller, which is complex; we needed to know if consumer mixed-reality hand and head tracking is a viable operator interface. Third, before any field application, the system had to be trainable in simulation, runnable on hardware, and documented so others can reproduce the procedure.

The first phase of my internship focused on engineering and replication: make the stack work reliably, not on deploying a specific real-world scenario. Only after teleoperation alignment was reasonably stable did my supervisor and I begin discussing paper framing and experiments. That application thread is summarized in Section 6.

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3. Starting Point (Prior Work in the Lab)

The baseline repository is visionpro-g1-inspire-teleop (seven commits, April–May 2026). At the start of my internship it already provided:

| Component | Capability |

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What was not yet in place: head-driven walking, a full MuJoCo sim workflow, staged calibration (F ->] -> S -> T), dual-hand drivers, keyboard-assisted walking for lab debugging, experiment logging, and consolidated English operations documentation.

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4. Timeline and Technical Work

4.1 Simulation workflow (early internship)

My supervisor first asked me to learn how the robot stack is operated and trained in simulation. I set up a three-terminal workflow: (1) MuJoCo simulation, (2) SONIC policy deploy, (3) Vision Pro bridge. One-click launch scripts (run_sonic_sim_loop.sh, run_sonic_deploy.sh, run_sonic_avp_teleop.sh) were added so the procedure does not depend on memorizing long command lines. This step established a safe place to practice before using the real robot.

4.2 Right-hand alignment (limited progress)

I initially spent time on right-hand alignment—improving how closely the robot right arm and Inspire hand follow the operator's hand in space and orientation. Progress was slower than expected because of mapping sensitivity, calibration drift, and coupling with the whole-body controller. Rather than blocking the rest of the project on perfect arm alignment, we prioritized head-driven locomotion so that whole-body teleoperation could move forward in parallel. Arm alignment tuning (especially the left arm) continued later as a separate track.

4.3 Head-driven locomotion

I implemented head-driven locomotion so the operator can walk the robot without using the hands for movement. The design separates two signals from the Vision Pro head pose:

- Head displacement (translation): horizontal motion of the head relative to a calibrated reference is converted into forward, backward, and lateral walking commands.

- Head fac

instantane
fixed head

The pipeline applies smoothing and velocity dead zones so small head jitter does not move the robot. When the operator lowers the head beyond a threshold, the system commands a lower pelvis height (squat/kneel band), which supports inspection in reduced vertical clearance. On the real robot, base orientation feedback is used to reduce gradual heading drift during corridor-style walking.

This capability was developed primarily to complete whole-body teleoperation; it later aligns with the REMS idea of keeping both hands free during transit, but that application framing came after the engineering work.

4.4 Staged calibration

I added a four-step calibration protocol so operator motion matches the robot more consistently:

1. F (CALIB_FULL) — Operator holds a forearms-forward L-shape for ~2 s; the system records reference head and wrist poses.

2.] — Enga

H (HEAD zero) re-zeros walking facing and squat height mid-session without repeating the full arm calibration. This protocol reduced sudden arm drops at policy start and improved repeatability between sessions.

4.5 Egocentric video (robot first-person view) — not yet complete

Infrastructure was added to stream the simulated head camera toward the Vision Pro via WebRTC (MuJoCo image publish on port 5555, bridge forwarding). However, this feature is not successfully usable today. Due to rendering / streaming issues, the operator cannot reliably see the robot's first-person view inside the Apple Vision Pro. The code path exists, but FPV should be treated as unfinished work, not a demonstrated result of this internship.

4.6 Real-robot stability and lab controls

On hardware I improved arm command smoothing, capped motion rates, and yaw stabilization for head locomotion. Real-robot launch requires --no-mujoco-fpv and the correct network interface. Optional keyboard overlay (--hybrid-locomotion: W/A/S/D) helps fine positioning in a small lab; default teleop remains head-only for consistency with the hands-free locomotion design.

4.7 Dual hands, packaging, and documentation

Support was extended for left and right Inspire hands (dual Modbus drivers and DDS topics). Scripts and logic were reorganized into the g1_teleop Python package with bin/ launchers and docs/OPERATIONS.md (sim vs. real robot, head-only vs. hybrid keyboard).

4.8 Parameter tuning (ongoing)

Substantial time went into making teleoperation feel controllable, not merely functional.

Left arm. The left arm is harder to align than the right because the left/right kinematic transforms differ. Raising the left hand sometimes caused torso twist or hunched posture, or insufficient reach height. Hand tracking loss previously caused large body motions when targets snapped to the init pose; I added hold-last-frame behavior when the wrist is temporarily lost.

Walking heading. Heading sometimes drifts (e.g., after backing up and walking forward again). Mitigations include pure head locomotion by default, H to re-zero facing, and optional keyboard hybrid on the real robot.

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5. Current Capabilities (Informal Testing)

These are informal lab observations, not formal study results.

Tasks that currently work reasonably well (mostly single-hand, basic manipulation):

Task	Notes

Tasks that remain difficult:

Task	Notes

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6. Application Framing and Experiments (After Alignment)

Once whole-body teleoperation was mostly working, my supervisor and I discussed how to position the work in a paper and what experiments are feasible. I have been part of these discussions throughout the second half of the internship.

We initially considered the REMS scenario in the draft PDF: an expert stays at a safe location and teleoperates a humanoid into hazardous confined spaces (tunnels, mines) for inspection and manipulation, signing off without first-entry exposure. That narrative matches our head locomotion and hands-free walking direction.

We then judged that full hazardous-environment claims would require capabilities the platform does not yet reliably support: rough terrain, heavy carrying, overhead looking, and complex bimanual loading. Rather than overclaiming, we narrowed the experimental plan to tasks the system can support today: basic single-hand manipulation and a simplified corridor walking study (Task A draft in docs/EXPERIMENT_TASK_A.md) with logged timing and heading error—not a claim of deployed underground operation.

Agreed paper-level story (draft):

- Contribution 1: Single Vision Pro, no extra hardware, for whole-body G1 teleoperation with a documented workflow.

- Contribut

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7. Summary

Completed during the internship

1. MuJoCo sim workflow and one-click scripts for operator training.

2. Head-dr

Not completed / in progress

- Egocentric video in Vision Pro (rendering issue; not demo-ready).

- Two-hand

Brief oral summary for a five-minute check-in

I am a Mitacs GRI intern reporting two to three months of work on teleoperating the full G1 humanoid with only an Apple Vision Pro, building on prior arm/hand code. I first learned the simulation workflow, worked on right-hand alignment with limited progress, then implemented head-driven walking and calibration so the body and arms can be controlled together. Robot first-person view in the headset is not working yet. After the stack was usable, my supervisor and I discussed REMS-style applications but scoped experiments down to single-hand tasks we can actually do (grasp, place in box, open door) while two-hand box carrying remains hard. Next steps are formal data collection on basic tasks and the corridor walking study.

Appendix: Commit Record (Optional)

Prior student (Apr–May 2026): Arm/hand teleop, SONIC bridge, finger mapping, real-robot tuning.

Author (Jul–Aug 2026): Sim workflow, head locomotion, staged calibration, FPV pipeline (incomplete UX), real-robot smoothing, IMU yaw assist, package refactor, hybrid keyboard, dual-hand drivers.

Author (Aug–Sep 2026, partly uncommitted): Left-arm parameters, arm tracking hold, Task A corridor draft, default teleop presets.

Prepared from Git history, lab notes, and supervisor discussions. Operator procedures: docs/OPERATIONS.md.